



# SIH26082 Technical Research Dossier

## Executive verdict

**Verdict: MODIFY** — proceed with a scientifically honest hybrid system.

Build a Delhi-NCR 72-hour forecasting platform combining:

1. CPCB/CAAQMS observations.
2. Operational numerical-weather forecasts.
3. ERA5 historical meteorology and pressure-level profiles.
4. NASA FIRMS fire detections and FRP.
5. CAMS/MERRA-2 chemical fields.
6. A calibrated pollutant-level ML model.
7. HYSPLIT or FLEXPART plume trajectories.
8. A limited, precomputed WRF-Chem demonstration showing genuine aerosol-meteorology feedback.
9. A GIS dashboard with uncertainty, inversion, PBL, fire and plume explanations.

Do not promise a fully operational, high-resolution WRF-Chem forecast running from scratch during a 36-hour hackathon. That is scientifically attractive but operationally risky. The strongest SIH strategy is **WRF-Chem-informed hybrid AI**, not “AI-only” and not an unvalidated claim of real-time WRF-Chem.

The official SIH portal was not directly accessible in the search results. The most complete available problem-statement copy identifies SIH26082 as a Ministry of Earth Sciences/NCMRWF software problem titled “Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus).” Treat the SIH mirror as a secondary copy and verify the final wording against the official portal before submission. [\[1\]](#) [\[2\]](#) [\[3\]](#)

## 1. What SIH expects

### Problem-statement interpretation

The available SIH26082 copy requires a high-resolution Delhi-NCR system that:

- Forecasts AQI for the next 72 hours.
- Couples weather and pollution rather than treating them as independent variables.
- Uses WRF-Chem or a similar open-source coupled framework.
- Models at least PM2.5 and ground-level ozone.
- Handles PM10, O3, NOx, temperature, wind and PBL height.
- Represents atmospheric inversion.
- Tracks external pollution episodes such as stubble burning.
- Predicts stubble-burning plume dispersion.
- Provides a real-time, user-friendly dashboard.
- Represents the feedback in which pollutants affect local meteorology.

The published copy explicitly describes aerosol effects on sunlight, temperature, wind and PBL height, and asks for inversion and stubble-plume features.<sup>[2]</sup> <sup>[3]</sup>

## Mandatory requirements

| Requirement                     | Interpretation                                                                           | Compliance test                                                      |
|---------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 72-hour forecast                | Forecast must extend to $t + 72$ hours, not only current AQI                             | Show hourly forecast through 72 hours                                |
| Delhi NCR focus                 | Delhi plus influential NCR and upwind regions                                            | Map and model domain must exceed Delhi municipal boundary            |
| High resolution                 | Finer than global products; exact required resolution must be confirmed from official PS | State grid resolution explicitly                                     |
| Coupled weather-pollution model | Meteorology and chemistry must interact                                                  | Show chemical transport and aerosol-radiation or equivalent feedback |
| Pollutants                      | PM2.5, PM10, O3 and NOx-related variables                                                | Forecast pollutant concentrations, not only AQI                      |
| Inversion                       | Diagnose and display trapping conditions                                                 | Show inversion metric and episode detection                          |
| Stubble burning                 | Detect fire activity and estimate transport influence                                    | Use satellite fire observations plus meteorological transport        |
| Dashboard                       | Real-time or near-real-time user-facing product                                          | Show live observations and forecast layers                           |
| Open-source framework           | WRF-Chem or scientifically comparable software                                           | Include reproducible model component                                 |

## Implicit technical requirements

The system also needs:

- Timestamp harmonisation in IST and UTC.
- Quality control for sensor data.
- Forecast initialisation from recent observations.
- Missing-data handling.
- Spatial interpolation from stations to grid.
- Meteorological forecast inputs.
- Chemical and emission inputs.
- Source-region representation.
- Model validation against independent stations.
- Forecast uncertainty.
- Reproducibility and provenance.
- Clear separation between observations, forecasts and derived products.
- A fallback mode if one upstream data source fails.

## Expected outputs

The minimum credible outputs are:

- Hourly PM2.5 forecast for 72 hours.
- Hourly PM10 forecast for 72 hours.
- Hourly O3 forecast for 72 hours.
- NO2 or NOx forecast if data quality permits.
- Derived Indian AQI.

- Temperature, humidity, wind, rainfall and PBL forecast.
- Inversion strength and duration.
- Fire hotspots and FRP.
- Upwind plume trajectory.
- Station-level and grid-level forecast maps.
- Actual-versus-predicted validation plots.
- Confidence interval or forecast range.
- Actionable alerts.

## Scientific requirements

A technically compliant system must distinguish:

- **Weather-only dispersion effects:** wind, rainfall, turbulence and PBL height.
- **Chemical transformation:** ozone formation and secondary aerosol.
- **Emission changes:** traffic, industry, dust and biomass burning.
- **Aerosol feedback:** aerosol radiation interaction affecting boundary-layer development.
- **Transport:** movement from Punjab, Haryana, Uttar Pradesh and Rajasthan into Delhi NCR.

## What is not compliant

The following would likely be challenged:

- Predicting AQI directly from historical AQI with no physical or meteorological component.
- Showing FIRMS fire points and calling them PM2.5 emissions.
- Calling a model “two-way coupled” when it only uses temperature and wind as ML inputs.
- Running HYSPLIT trajectories and calling that chemical feedback.
- Claiming “high resolution” while presenting only a 40-km or 0.25° global grid.
- Presenting a static historical animation as live forecasting without labelling it.
- Reporting only Delhi city-average AQI.
- Claiming exact sectoral source attribution without tagged chemistry or source-apportionment evidence.
- Using a direct AQI prediction that cannot explain pollutant-level exceedances.
- Showing a WRF-Chem screenshot without describing configuration, emissions and validation.

# 2. Delhi-NCR scientific system

## Meteorological controls

Delhi pollution is strongly controlled by:

- **Wind speed:** low wind reduces horizontal ventilation.
- **Wind direction:** determines whether air arrives from Punjab/Haryana, Uttar Pradesh, Rajasthan or cleaner sectors.
- **PBL height:** a shallow boundary layer confines emissions into a smaller volume.
- **Temperature inversion:** warmer air above cooler surface air suppresses vertical mixing.
- **Humidity:** promotes hygroscopic particle growth and aqueous-phase chemistry.
- **Rainfall:** removes soluble particles through wet scavenging.

- **Radiation:** drives ozone photochemistry and affects surface heating.
- **Cloud cover:** reduces radiation and changes photochemistry and boundary-layer growth.
- **Turbulence and stability:** determine vertical dilution.
- **Surface pressure and synoptic circulation:** influence regional transport and stagnation.

A simplified near-surface concentration relationship is:

$$C \propto \frac{E_{\text{local}} + E_{\text{transported}}}{H_{\text{PBL}} U}$$

where  $C$  is concentration,  $E$  is emission or transported mass,  $H_{\text{PBL}}$  is mixing height and  $U$  is ventilation-related wind speed. This is a conceptual relationship, not a substitute for a chemical transport model.

## Chemistry

### PM2.5

PM2.5 includes:

- Primary organic aerosol.
- Black carbon.
- Sulfate.
- Nitrate.
- Ammonium.
- Secondary organic aerosol.
- Fine dust and combustion ash.

Delhi's severe episodes are often dominated by a mixture of local emissions, transported particles and secondary aerosol. A recent WRF-Chem study of northern India found Delhi PM2.5 contributions from nitrate, dust, secondary organic aerosol and ammonium, while also finding that model errors were linked to emissions and chemistry uncertainty.<sup>[4]</sup>

### PM10

PM10 includes PM2.5 plus coarse material such as:

- Road dust.
- Construction dust.
- Soil and mineral particles.
- Mechanical resuspension.
- Larger combustion particles.

It is especially sensitive to wind-driven dust and construction activity.

## Ozone

Ground-level O<sub>3</sub> is not emitted directly in the same way as PM. It forms through photochemical reactions involving:

- NO<sub>x</sub>.
- VOCs.
- Sunlight.
- Temperature.

- Transport.
- Radical chemistry.

High particulate loading can reduce surface radiation and alter photochemistry, while the response of O<sub>3</sub> depends on chemical regime, aerosol optical properties and vertical mixing.

### **NO, NO<sub>2</sub> and NO<sub>x</sub>**

- NO is emitted mainly by combustion and rapidly reacts with ozone.
- NO<sub>2</sub> participates in ozone formation and nitrate aerosol chemistry.
- NO<sub>x</sub> is the combined family commonly approximated as NO + NO<sub>2</sub>.

NO<sub>x</sub> is useful both as a forecast target and as an indicator of traffic and combustion influence, but interpreting it as a direct source marker requires caution.

### **SO<sub>2</sub>, CO and VOCs**

Include SO<sub>2</sub>, CO and VOC proxies where available:

- SO<sub>2</sub> helps represent industrial and sulfur chemistry.
- CO is a useful combustion and biomass-burning tracer.
- VOCs are essential for realistic ozone chemistry, but routine spatial coverage is limited.

### **Two-way coupling**

#### **Pollution to weather**

The sequence is:

1. Aerosols scatter and absorb solar radiation.
2. Less shortwave radiation reaches the surface.
3. Surface heating and sensible heat flux decrease.
4. The boundary layer grows more slowly or collapses.
5. Turbulent mixing weakens.
6. Pollution remains concentrated near the surface.
7. Higher aerosol loading can further modify radiation and stability.

Black carbon can additionally heat elevated layers, producing vertical temperature changes that may strengthen or alter stability.

#### **Weather to pollution**

The reverse sequence is:

1. Wind transports and dilutes pollutants.
2. PBL height controls the volume available for mixing.
3. Inversion suppresses vertical exchange.
4. Rain removes particles.
5. Humidity changes particle size and chemical production.
6. Sunlight and temperature affect ozone chemistry.
7. Synoptic patterns control regional inflow.

## Genuine two-way coupling

A scientifically defensible two-way implementation must allow aerosol or chemical variables to modify meteorological calculations, normally through:

- Aerosol-radiation interaction.
- Aerosol-cloud interaction, if enabled and validated.
- Photolysis changes.
- Feedback on temperature, surface energy balance and boundary-layer dynamics.

WRF-Chem uses shared meteorological and chemical coordinates and can represent feedback through radiation and photolysis. <sup>[5]</sup> <sup>[4]</sup>

## What does not qualify

These are one-way or post-processing systems:

- Feeding weather variables into XGBoost.
- Running WRF weather and separately plotting pollution.
- Running HYSPLIT trajectories without chemistry or feedback.
- Adding a manually designed “pollution correction” to PBL height.
- Displaying aerosol optical depth without feeding aerosols into radiation.
- Using an AI model whose weather inputs are only explanatory features.

For the SIH prototype, label the system accurately:

“Operational hybrid forecast with WRF-Chem feedback-informed calibration and an offline WRF-Chem coupled demonstration.”

Only call the WRF-Chem experiment itself “two-way coupled.”

# 3. Geographic influence model

## Formal NCR scope

Delhi NCR includes:

- The entire National Capital Territory of Delhi.
- NCR districts in Haryana.
- NCR districts in Uttar Pradesh.
- NCR districts in Rajasthan.

The exact current district list and boundary should be loaded from the NCR Planning Board or an authoritative government boundary source before final submission. District reorganisations and boundary definitions can change, so do not hard-code a list copied from an old paper.

## Influential regions beyond NCR

Important surrounding regions include:

- Punjab agricultural districts.
- Haryana agricultural districts.
- Western Uttar Pradesh.
- Parts of Rajasthan.

- The wider Indo-Gangetic Plain.
- Himalayan foothill and western disturbance-influenced regions.
- Industrial corridors and urban clusters outside NCR.

Delhi cannot be treated as a closed box because the residence time of air parcels can be longer than the time needed for regional pollution to travel into the city.

### Seasonal transport

| Season       | Typical controls                                                       | Likely influence                                                         |
|--------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Winter       | Weak winds, shallow PBL, frequent inversion, fog and stable conditions | Local accumulation plus regional transport; strongest severe-smog season |
| Post-monsoon | Crop-residue burning, changing winds, dry conditions                   | Punjab/Haryana smoke can reach Delhi under suitable flow                 |
| Monsoon      | Strong mixing and rainfall                                             | Wet scavenging and dilution; generally lower particulate burden          |
| Summer       | Strong convection, dust, higher mixing                                 | Dust and heat; O <sub>3</sub> can be important                           |
| Pre-monsoon  | Dryness, dust, thunderstorms and changing winds                        | Dust episodes, convective transport and photochemistry                   |

### Source-to-receptor representation

Use a dynamic influence graph:

```

Punjab/Haryana fires
  ↓
Fire emissions + FRP
  ↓
Wind field + PBL + stability
  ↓
Transport corridor / plume
  ↓
NCR boundary inflow
  ↓
Delhi station and grid concentrations
  ↓
Downwind Uttar Pradesh / Rajasthan

```

The source-region contribution should be expressed as:

- “Transport-favourable.”
- “Likely influence.”
- “Modelled contribution under scenario.”
- “Fire-enhanced risk.”

Do not state exact percentage contributions unless produced by tagged chemistry or a defensible source-apportionment experiment.

### Primary demonstration period

Use **October–November**, particularly a severe post-monsoon pollution episode, because it simultaneously demonstrates:

- Stubble-burning activity.
- Low PBL conditions.
- Regional transport.
- Inversion.

- High PM2.5.
- Visible plume movement.
- Policy relevance.

A strong historical case is **6–13 November 2017**, for which a WRF-Chem study specifically evaluated 72-hour forecasts, MODIS AOD assimilation, real-time fire emissions and aerosol-radiation interactions.<sup>[6]</sup>

A second case is **30 October–7 November 2016**, examined in a recent northern-India WRF-Chem evaluation.<sup>[4]</sup>

A third candidate is the **November 2016 severe Delhi episode**, during which reported average PM2.5 increased from 142 µg/m<sup>3</sup> before the event to 563 µg/m<sup>3</sup> during it.<sup>[7]</sup>

## 4. Monitoring network

### Authoritative source

Use CPCB's Central Control Room and CAAQMS station list as the authoritative station registry. CPCB provides an All-India station PDF with station name, state, city, operating agency, station code and coordinates. The available listing showed Delhi with 39 stations in one snapshot, but the network changes over time and different sources may count active, inactive, legacy or duplicate stations differently.<sup>[8] [9]</sup>

CPCB also provides a Delhi-NCR station list and real-time monitoring interface.<sup>[10] [11]</sup>

### Why a fixed hand-written inventory is unsafe

A complete station inventory should be generated automatically on the data-refresh date because:

- Stations are added or retired.
- Operators change.
- Station names may be duplicated.
- Agencies may publish different station registries.
- A station can appear in the registry but have incomplete current data.
- Pollutant availability differs by station.
- Station coordinates and station codes may be corrected.

Therefore, create a versioned stations table from the current CPCB registry and retain the retrieval date.

### Required station inventory fields

```
station_id
station_name
operator
city
district
state
latitude
longitude
station_code
station_status
monitoring_type
pollutants_available
pm25_available
pm10_available
o3_available
no_available
no2_available
nox_available
so2_available
```



```
co_available
sampling_interval
first_observation
last_observation
real_time_status
data_url
validation_role
site_class
registry_version
retrieved_at
```

## Station grouping

### Delhi

Include all CPCB/DPCC/IMD/IITM-listed CAAQMS stations, including examples such as:

- Alipur.
- Shadipur.
- IHBAS Dilshad Garden.
- NSIT Dwarka.
- DTU.
- ITO.
- Sirifort.
- Mandir Marg.
- Anand Vihar.
- R.K. Puram.
- Punjabi Bagh.
- Aya Nagar.
- Lodhi Road.
- North Campus.
- Burari Crossing.
- CRRRI Mathura Road.
- Pusa.
- IGI Airport.
- East Arjun Nagar.
- Ashok Vihar.
- Jawaharlal Nehru Stadium.
- Nehru Nagar.
- Dwarka Sector 8.
- Patparganj.
- Sonia Vihar.
- Jahangirpuri.
- Rohini.
- Najafgarh.
- Vivek Vihar.

- Narela.
- Okhla Phase 2.
- Wazirpur.
- Bawana.
- Sri Aurobindo Marg.

This is illustrative, not a substitute for downloading the latest registry. The CPCB technical document lists many of these stations and agencies. <sup>[12]</sup>

## **Haryana NCR**

Prioritise stations in:

- Gurugram.
- Faridabad.
- Sonipat.
- Panipat.
- Rohtak.
- Jhajjar.
- Rewari.
- Palwal.
- Nuh.
- Bhiwani where included by the current NCR definition.

## **Uttar Pradesh NCR**

Prioritise:

- Noida.
- Greater Noida.
- Ghaziabad.
- Hapur.
- Meerut.
- Bulandshahr.
- Baghpat.
- Muzaffarnagar where relevant to the current boundary and transport analysis.

## **Rajasthan NCR**

Prioritise:

- Alwar.
- Bharatpur.
- Other current NCR districts listed by the NCR Planning Board.

Outside-NCR upwind network

Use stations and regional gridded fields in:

- Punjab.
- Haryana outside NCR.
- Western Uttar Pradesh.
- Rajasthan.
- The wider Indo-Gangetic Plain.

These stations are important for boundary conditions and transport diagnosis, even if not used as Delhi validation stations.

Station roles

| Role                   | Preferred stations                                        |
|------------------------|-----------------------------------------------------------|
| Training               | Long-duration stations with stable PM2.5 and PM10 records |
| Independent validation | Stations held out spatially and temporally                |
| Urban background       | Lodhi Road, Pusa-type locations                           |
| Traffic                | ITO, Anand Vihar, roadside stations                       |
| Industrial             | Bawana, Wazirpur and industrial-area stations             |
| Regional inflow        | Peripheral NCR and upwind stations                        |
| Event detection        | Stations with complete hourly PM2.5 and PM10              |
| Chemical validation    | Stations with O3, NO2, CO and SO2                         |

Do not classify a station as traffic, industrial or background only from its name. Use CPCB/DPCC metadata, surrounding land use, road density and a manually reviewed GIS classification.

5. Dataset matrix

Core datasets

| Dataset            | Organization                      | Variables                                                                 | Resolution                    | Coverage/access                                                                    | Use                                     |
|--------------------|-----------------------------------|---------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------|-----------------------------------------|
| CAAQMS             | CPCB/DPCC/IMD/IITM/state agencies | PM2.5, PM10, O3, NO2, NOx, SO2, CO, NH3 where available                   | Station/hourly or near-hourly | Real-time and selected historical downloads; portal workflows may change           | Targets, validation, bias correction    |
| CPCB AQI           | CPCB                              | AQI and sub-index inputs                                                  | Station/time                  | Web dashboard and data resources                                                   | Verification and user-facing comparison |
| OGD AQI resource   | Government of India/CPCB          | Real-time AQI                                                             | Station/time                  | <a href="#">Data.gov.in</a> resource; API access depends on resource configuration | Backup/metadata                         |
| ERA5 single levels | ECMWF/Copernicus                  | 2-m temperature, dew point, wind, pressure, radiation, precipitation, BLH | 0.25°, hourly                 | 1940–present; approximately 5-day operational latency                              | Historical features and hindcasts       |

| Dataset                 | Organization                         | Variables                                        | Resolution                                                      | Coverage/access                                       | Use                                                 |
|-------------------------|--------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|
| ERA5 pressure levels    | ECMWF/Copernicus                     | Vertical temperature, humidity and wind          | 0.25°, hourly/selected levels                                   | Long historical archive                               | Inversion and stability                             |
| GFS                     | NOAA                                 | Forecast weather fields                          | Global forecast grid; version-dependent                         | Operational forecast                                  | Near-real-time weather input                        |
| CAMS global composition | Copernicus/ECMWF                     | PM, O3, NO2, aerosols and gases                  | Approximately 0.4°, hourly single-level or 3-hourly multi-level | Historical archive and forecast products              | Chemical background, ensemble feature               |
| MERRA-2                 | NASA                                 | Aerosols, meteorology, composition               | Approximately 0.5° × 0.625°                                     | Historical reanalysis                                 | Aerosol evaluation and fallback                     |
| FIRMS MODIS/VIIRS       | NASA                                 | Fire location, confidence, FRP                   | MODIS 1 km; VIIRS 375 m pixel detections                        | NRT and archive; API registration/token may be needed | Fire features and plume source locations            |
| MODIS AOD               | NASA                                 | Aerosol optical depth                            | Satellite swath/product-dependent                               | Historical; cloud gaps                                | Aerosol assimilation/evaluation                     |
| TROPOMI/Sentinel-5P     | ESA/Copernicus                       | NO2, CO, O3, SO2, aerosol-related products       | Kilometre-scale satellite pixels, product-dependent             | Historical and near-real-time                         | Column-composition features and regional patterns   |
| OpenStreetMap           | OSM contributors                     | Roads, industries, land use proxies              | Vector                                                          | Download/API; ODbL obligations                        | Static emissions proxies and map layers             |
| GADM/Natural Earth      | Open geographic data                 | Administrative boundaries                        | Vector                                                          | Downloadable                                          | Boundaries and aggregation                          |
| NCR boundary            | NCR Planning Board/government source | NCR and district boundaries                      | Vector                                                          | Verify current official source                        | Domain clipping and district views                  |
| EDGAR/HTAP/CEDS         | Research/government inventories      | Sectoral emissions                               | Gridded; product-dependent                                      | Historical; licensing varies                          | Background emissions and prior                      |
| GFED/FINN/GFAS          | Research/NASA/NCAR                   | Biomass-burning emissions or fire radiative data | Product-dependent                                               | Research downloads; access varies                     | Fire emission estimates and sensitivity experiments |

ERA5 provides hourly global reanalysis from 1940 onward at 0.25° atmospheric resolution, with pressure-level data useful for vertical profiles. CAMS provides global atmospheric-composition forecasts and analyses with many chemical species and approximately 40-km-scale products, depending on product/version. <sup>[13] [14] [15] [16]</sup>

NASA FIRMS provides MODIS and VIIRS active-fire data in formats such as CSV, KML and shapefile; VIIRS active-fire detections represent the centre of a 375-m pixel, while FRP is reported in megawatts. <sup>[17] [18] [19] [20]</sup>

## Data-availability reality check

| Source                          | Status | Main limitation                                                                                     |
|---------------------------------|--------|-----------------------------------------------------------------------------------------------------|
| CPCB current AQI/CAAQMS         | ☒      | Current access is practical, but bulk historical retrieval and portal endpoints can be inconsistent |
| <a href="#">Data.gov.in</a> AQI | ☒      | Resource/API configuration must be checked for the specific dataset                                 |
| ERA5                            | ☒      | Registration and CDS credentials required; not real-time                                            |
| GFS                             | ☒      | Good operational fallback; coarse relative to urban modelling                                       |
| CAMS                            | ☒/☒    | Strong chemical background product, but coarse for NCR streets and access policies vary             |

| Source                    | Status | Main limitation                                                                                                            |
|---------------------------|--------|----------------------------------------------------------------------------------------------------------------------------|
| FIRMS NRT                 | ☒      | Excellent for fire detection; cloud, overpass and omission limitations                                                     |
| FIRMS archive             | ☒      | Registration, product selection and archive workflow required                                                              |
| MODIS AOD                 | ☒      | Cloud gaps, retrieval uncertainty and column-to-surface conversion problem                                                 |
| TROPOMI                   | ☒/☒    | Strong regional trace-gas information; satellite overpass and cloud limitations                                            |
| IMD station/forecast data | ☒      | Institutional access and redistribution conditions may apply                                                               |
| SAFAR data                | ☒      | System exists and provides 1–3-day information, but openly downloadable, station-level historical access is not guaranteed |
| EDGAR/HTAP/CEDS           | ☒      | Suitable for priors, not local real-time emissions                                                                         |
| Indian state inventories  | ☒      | Availability and licensing vary substantially                                                                              |
| OpenAQ                    | ☒/☒    | Useful aggregator, but source-level licensing and provenance must be tracked                                               |
| HYSPLIT                   | ☒      | Access/use conditions apply; NOAA agreement restricts redistribution of software                                           |
| FLEXPART                  | ☒      | GPL software, but meteorological preprocessing and operation are non-trivial                                               |
| WRF-Chem                  | ☒      | Open research software, but configuration, compilation and compute are difficult                                           |

CPCB’s published technical material confirms that CAAQMS data are made available through the CPCB control-room ecosystem, while its current station list identifies operators and station codes. <sup>[21]</sup> <sup>[8]</sup> <sup>[12]</sup>

Do not claim that every portal has an unrestricted API. For every endpoint, record:

```
retrieval_date
endpoint
authentication_required
terms_checked
response_schema
rate_limit
historical_depth
```

## 6. Master dataset design

### Recommended spatial grid

Use two resolutions:

1. **Operational grid:** 0.05° to 0.1° for Delhi NCR.
2. **Station layer:** raw station points for validation.
3. **Plume layer:** vector trajectories and raster influence fields.
4. **Optional research grid:** 1–3 km if a WRF-Chem nested domain is available.

A 0.05° grid is approximately 5 km and is realistic for a student prototype. Calling it “street-level” or “hyperlocal” would be incorrect.

### Time resolution

Use:

- Raw ingestion: 5-minute, 15-minute or hourly depending on source.
- Canonical modelling interval: hourly.
- Forecast: hourly  $t + 1$  through  $t + 72$ .
- Dashboard aggregation: hourly and 3-hourly display options.

## Example schema

```
timestamp_utc
timestamp_ist
forecast_issue_time
lead_hour
grid_id
latitude
longitude
station_id
district
state
pm25_obs
pm10_obs
o3_obs
no_obs
no2_obs
nox_obs
so2_obs
co_obs
temperature_2m
relative_humidity
dewpoint_2m
surface_pressure
u10
v10
wind_speed
wind_direction
precipitation
cloud_cover
shortwave_radiation
pbl_height
geopotential_height_925
geopotential_height_850
temperature_925
temperature_850
inversion_strength
inversion_layer_base
inversion_layer_top
inversion_duration
aod_550
cams_pm25
cams_o3
fire_count_24h
fire_count_48h
fire_count_72h
fire_frp_sum
fire_frp_weighted
upwind_fire_count
upwind_frp
transport_sector
trajectory_arrival_probability
dust_proxy
road_density
industrial_density
population_density
land_use_class
weekday
hour_local
day_of_year
pm25_lag_1h
pm25_lag_3h
pm25_lag_6h
pm25_lag_24h
pm25_rolling_6h
pm25_rolling_24h
```

```
target_pm25_t1
target_pm25_t6
target_pm25_t24
target_pm25_t48
target_pm25_t72
data_quality_flag
source_version
```

Do not put observations that were unavailable at forecast issue time into training features for that issue. Every row needs an `as_of_time` or `forecast_issue_time`.

## 7. Preprocessing pipeline

### Pipeline

```
Source ingestion
↓
Raw immutable storage
↓
Schema validation
↓
Unit and timezone normalisation
↓
Duplicate removal
↓
Sensor quality control
↓
Temporal resampling
↓
Spatial joining/interpolation
↓
Derived meteorology
↓
Inversion/PBL calculation
↓
Fire and plume feature generation
↓
Feature store
↓
Leakage-safe train/validation split
```

### Quality-control rules

- Reject impossible values such as negative PM or humidity above 100%.
- Flag, rather than automatically delete, extreme values.
- Detect flat-line sensors.
- Detect sudden step changes.
- Identify repeated timestamps.
- Preserve original and cleaned values.
- Apply station-specific missingness statistics.
- Require minimum completeness for daily and rolling features.
- Maintain a `quality_flag` column.
- Avoid replacing long missing periods with interpolation.

## Wind encoding

Use:

$$u = s \sin(\theta), \quad v = s \cos(\theta)$$

with a convention explicitly documented for meteorological direction. Wind direction should not be used as raw degrees because  $0^\circ$  and  $360^\circ$  are adjacent.

## Spatial interpolation

Do not blindly interpolate every pollutant using ordinary kriging.

Use:

- IDW or kriging for exploratory maps.
- Land-use regression or graph-based interpolation for station-to-grid estimates.
- Model background fields from CAMS/MERRA-2.
- Station residual correction for local bias.
- Separate validation stations from interpolation stations.

A practical field estimate is:

$$\begin{aligned} & \hat{C}_{\text{grid}} \\ &= C_{\text{background}} \\ &+ \text{Interp} \\ & \left( C_{\text{station}} - C_{\text{background, station}} \right) \end{aligned}$$

This is usually more defensible than interpolating raw PM across large distances.

## Leakage prevention

For each forecast issue time  $t_0$ :

- Use only observations with timestamp  $\leq t_0$ .
- Use only weather forecast values available at  $t_0$ .
- Use fire observations whose publication latency would permit availability at  $t_0$ .
- Do not use revised ERA5 values as if they were operational forecasts.
- Do not use future rolling averages.
- Split validation by date, not random rows.
- Use blocked temporal cross-validation.
- Hold out entire stations for spatial generalisation tests.
- Maintain an “operational simulation” dataset with realistic latency.



## 8. Inversion and PBL

### Inversion definition

An inversion exists where temperature increases with height over a layer:

$$\frac{\partial T}{\partial z} > 0$$

A surface inversion occurs when the near-surface layer is cooler than air above it.

### Feasible methods

| Method                         | Data                                | Strength                  | Limitation                                |
|--------------------------------|-------------------------------------|---------------------------|-------------------------------------------|
| ERA5 pressure-level lapse rate | Vertical temperature                | Easy and reproducible     | Coarse vertical and horizontal resolution |
| WRF vertical profile           | High-resolution model profile       | Best for model domain     | Dependent on WRF quality                  |
| Radiosonde                     | Observed vertical temperature       | Strong validation         | Sparse in time/location                   |
| Surface proxy                  | Night cooling, dew point, wind, PBL | Operationally easy        | Not a true inversion measurement          |
| Richardson number              | Wind shear and stratification       | Scientifically meaningful | Requires reliable vertical profiles       |
| PBL threshold                  | ERA5/WRF BLH                        | Useful operational proxy  | PBL height is model-dependent             |

### Recommended Inversion Strength Index

Use a transparent index rather than inventing universal physical categories:

$$ISI = w_1 \max(0, \Delta T_{925-1000}) + w_2 \max(0, \Delta T_{850-925}) + w_3 f(H_{\text{PBL}}) + w_4 f(U_{10}) + w_5 f(\text{stability})$$

where:

- $\Delta T$  represents temperature increase with altitude.
  - $H_{\text{PBL}}$  is PBL height.
  - $U_{10}$  is 10-m wind speed.
  - $f$  functions are normalised using historical Delhi percentiles.

Call the outputs:

- Low trapping potential: below the 25th percentile.
- Moderate: 25th–75th percentile.
- High: 75th–90th percentile.
- Extreme: above the 90th percentile.

These are **operational risk categories**, not universal atmospheric classifications. Calibrate them against radiosonde observations and observed PM accumulation.

### Dashboard presentation

Show:

- Inversion score.
- Temperature profile.
- Estimated layer height.
- Duration.
- PBL height.

- Wind speed.
- “Trapping potential” language.

Avoid saying “severe inversion” without displaying the definition.

### PBL recommendation

Use:

- ERA5 PBL height for historical training.
- GFS or WRF PBL height for operational forecast.
- WRF-Chem PBL height for the coupled research case.
- CPCB PM2.5 as the target to learn a station-corrected PBL–pollution relationship.

ERA5 includes boundary-layer height among its single-level variables. <sup>[14]</sup>

## 9. Stubble burning and plume modelling

### Fire detection

Use FIRMS:

- VIIRS 375-m NRT detections for higher spatial detail.
- MODIS for continuity and historical analysis.
- Fire confidence.
- Acquisition time.
- FRP.
- Sensor and scan angle.
- Cloud and missing-observation flags.

FIRMS provides active-fire data globally with minimal delay, while the precise latency and data product differ by sensor and region. <sup>[19] [22] [17]</sup>

### Fire feature engineering

Create:

```
fire_count_24h
fire_count_48h
fire_count_72h
frp_sum_24h
frp_mean_24h
frp_max_24h
confidence_weighted_frp
distance_to_nor
upwind_fire_count
upwind_frp
wind_aligned_frp
fire_density_by_source_region
fire_age_hours
fire_cluster_id
```

Use a crop-calendar multiplier to distinguish likely paddy-residue burning from other fires.

A simple source-region influence score is:

\$\$

$$F_r(t) = \sum_{i \in r} FRP_i \cdot q_i$$

$$\begin{aligned} & \cdot e^{-\lambda \tau_i} \\ & \cdot \cos(\Delta \theta_i) \\ & \cdot \mathbb{1}(\text{upwind}) \end{aligned}$$

\$\$

where  $q_i$  is detection confidence,  $\tau_i$  is fire age and  $\Delta \theta_i$  measures alignment with the wind direction.

### Why FIRMS is not an emission inventory

A hotspot:

- Does not measure burned area exactly.
- May miss fires due to cloud or overpass timing.
- May detect the same fire in multiple observations.
- Does not provide fuel load directly.
- Does not directly measure PM2.5 released.
- Does not determine how much reaches Delhi.

Use FRP as an activity/intensity proxy. Convert it to emissions only with an emissions model containing:

- Fuel load.
- Combustion completeness.
- Emission factors.
- Fuel type.
- Area burned.
- Temporal allocation.
- Plume rise.

Label the output “fire-derived emission proxy” unless using a validated inventory.

### Plume model comparison

| Method                        | Scientific strength           | Compute       | Real-time            | Recommendation                                   |
|-------------------------------|-------------------------------|---------------|----------------------|--------------------------------------------------|
| Gaussian plume                | Low–medium                    | Low           | Excellent            | Local illustrative plume only                    |
| HYSPLIT trajectory            | Medium for transport          | Low–medium    | Good                 | Strong demo and trajectory layer                 |
| HYSPLIT dispersion            | Medium                        | Medium        | Good                 | Useful if configured carefully                   |
| FLEXPART                      | High for Lagrangian transport | Medium–high   | Good with automation | Strong advanced option                           |
| WRF-Chem                      | High for coupled chemistry    | Very high     | Difficult            | Research hindcast and selected case              |
| WRF + chemistry postprocessor | Medium–high                   | High          | Moderate             | Practical intermediate                           |
| ML plume model                | Variable                      | Low inference | Excellent            | Use only after physical features are established |

HYSPLIT is designed for trajectories, dispersion, transport, chemical transformation and deposition, but NOAA’s use agreement restricts redistribution of the software. FLEXPART is free software under GPLv3 and supports Lagrangian transport, deposition and related processes. <sup>[23] [24] [25] [26] [27]</sup>

## Recommendation

Use:

- **HYSPLIT** or **FLEXPART** for operational trajectory/influence maps.
- A **Gaussian** or **advection-diffusion approximation** for fast visual plume polygons.
- **WRF-Chem** for one or two historical episodes with explicit aerosol-radiation feedback.
- Fire-derived features in ML for pollutant correction.
- No exact source-percentage claim unless running tagged tracers or a validated source-apportionment model.

## 10. WRF-Chem feasibility

### What WRF-Chem provides

WRF-Chem integrates:

- Numerical weather prediction.
- Gas-phase chemistry.
- Aerosol chemistry.
- Emissions.
- Transport.
- Deposition.
- Radiation interactions.
- Photolysis.
- Nested domains.
- Vertical and horizontal chemical transport on the meteorological grid.

The WRF-Chem user guide recommends familiarity with WRF compilation and configuration before attempting chemistry simulations. <sup>[28]</sup>

### Required inputs

#### Meteorology

- Initial and boundary conditions.
- Surface fields.
- Land use.
- Soil data.
- Sea-surface temperature where relevant.
- Terrain.
- Radiation and microphysics configuration.

#### Chemistry

- Chemical initial and boundary conditions.
- Gas and aerosol emissions.
- Anthropogenic inventories.
- Biogenic emissions.

- Fire emissions.
- Dust emissions.
- Chemical mechanism.
- Aerosol module.
- Photolysis configuration.

## Recommended mechanisms

For the prototype:

- MOZART gas chemistry.
- MOSAIC aerosol mechanism if compute permits.
- A simpler chemistry configuration for debugging.
- Explicit aerosol-radiation feedback for the research case.

A 2024 India WRF-Chem study used ERA5 nudging, evaluated 2016 northern-India pollution, and included fire emissions and multiple aerosol comparisons. It found good meteorological skill but meaningful PM<sub>2.5</sub> biases, illustrating that model configuration and emissions remain major uncertainties. <sup>[4]</sup>

## Domain design

A credible nested setup:

- D01: South Asia or northern India, approximately 15–30 km.
- D02: Indo-Gangetic Plain, approximately 5–10 km.
- D03: Delhi NCR, approximately 1–3 km for a research case.

For a hackathon, do not promise D03 operationally unless you have already tested runtime and output quality.

## WRF-Chem weaknesses

- Long setup time.
- Compilation failures.
- Complex emissions preprocessing.
- Chemical mechanism sensitivity.
- High storage and compute requirements.
- Poor results if emissions are wrong.
- Boundary-condition and plume-rise sensitivity.
- Need for specialist interpretation.
- Forecast cycles can fail operationally.

WRF-Chem studies over Delhi have shown that even advanced configurations can underestimate peak PM<sub>2.5</sub> or overestimate dust, and that night-time wind and chemistry errors materially affect forecasts. <sup>[4]</sup>

## Practical architecture choice

Use WRF-Chem in three modes:

1. **Offline hindcast** for a historical severe episode.
2. **Sensitivity experiment** with aerosol-radiation feedback on/off.
3. **Feature generator/calibration source** for ML and dashboard explanations.

Run operational forecast with:

- GFS weather.
- CAMS/MERRA-2 chemical background.
- CPCB observations.
- FIRMS fire features.
- Plume trajectories.
- ML residual correction.

## 11. Architecture comparison

| Architecture                                   | Scientific compliance | Accuracy potential | Compute | Real-time | SIH feasibility | Verdict                    |
|------------------------------------------------|-----------------------|--------------------|---------|-----------|-----------------|----------------------------|
| A. Full operational WRF-Chem                   | 10                    | 8                  | 2       | 3         | 4               | Too risky as sole system   |
| B. WRF-Chem hindcast + AI correction           | 9                     | 9                  | 5       | 7         | 9               | Best scientific compromise |
| C. WRF + separate chemistry/plume + AI         | 7                     | 8                  | 6       | 8         | 8               | Good fallback              |
| D. Physics-guided AI using WRF/ERA5/CPCB/FIRMS | 6–7                   | 7–8                | 9       | 9         | 10              | Best operational MVP       |
| E. Hybrid multi-model ensemble                 | 9                     | 9                  | 6       | 8         | 9               | Recommended final design   |

### Recommended architecture

#### Offline scientific layer

- WRF-Chem historical case.
- Aerosol-radiation feedback on/off experiment.
- Fire-emission sensitivity.
- PBL and inversion comparison.
- WRF-Chem output used as a physics reference.

#### Operational layer

- GFS or approved operational meteorology.
- ERA5 for historical training.
- CAMS chemical background.
- CPCB observations.
- FIRMS fire features.
- HYSPLIT/FLEXPART trajectories.
- LightGBM/XGBoost station residual correction.
- Spatio-temporal neural model only if data volume supports it.

#### Model output

\$\$

$$\hat{Y}_{t+h}$$

$Y^{\text{physical/background}}_{t+h}$

+

$g^{\theta}$

$\left($

observations,

weather,

fire,

plume,

inversion,

spatial context

$\right)$

\$\$

This is more defensible than pure black-box forecasting and more feasible than full WRF-Chem operations.

## 12. ML model selection

### Model comparison

| Model              | Strength                                    | Weakness                                      | Best use               |
|--------------------|---------------------------------------------|-----------------------------------------------|------------------------|
| Persistence        | Essential baseline                          | No physical skill beyond short horizon        | Baseline               |
| Historical average | Seasonal reference                          | Poor event response                           | Baseline               |
| Random Forest      | Robust, easy                                | Weak extrapolation and spatial structure      | Baseline               |
| XGBoost            | Strong tabular performance                  | Separate models may be needed by horizon      | First serious model    |
| LightGBM           | Fast and scalable                           | Sensitive to leakage and tuning               | Main operational model |
| CatBoost           | Handles categorical features well           | Less necessary for all-numeric weather data   | Station-aware baseline |
| LSTM/GRU           | Temporal sequence learning                  | Can miss spatial transport and regime changes | Sequence baseline      |
| TCN                | Efficient long temporal receptive field     | Less familiar to judges                       | Strong sequence model  |
| CNN-LSTM/ConvLSTM  | Spatial-temporal grids                      | Requires dense gridded labels                 | Grid experiments       |
| Transformer/TFT    | Multi-horizon and interpretability features | Data and tuning hungry                        | Advanced model         |
| GNN/ST-GNN         | Station-network transport relationships     | Requires graph design and sufficient data     | Best advanced option   |
| PINN/Neural ODE    | Physics constraints                         | Difficult and not automatically superior      | Research only          |
| Ensemble           | Better robustness and uncertainty           | More engineering                              | Final forecast         |

### Recommended progression

1. Persistence.
2. Seasonal mean.
3. Weather-only LightGBM.
4. Pollution + weather LightGBM.
5. Fire/plume/inversion LightGBM.
6. ST-GNN or TCN if enough station history.
7. Ensemble with quantile outputs.

## Input structure

For station  $i$ , issue time  $t_0$ , lead  $h$ :

```
past 72-168 hours:
  PM2.5, PM10, O3, NO2, NOx
  weather
  PBL and inversion
  fire/plume features

future 72 hours:
  forecast weather
  forecast PBL
  CAMS background composition
  expected transport features

static:
  coordinates
  roads
  land use
  industrial density
  population
  station type
```

## Output structure

Use multi-output pollutant forecasts:

```
PM2.5[t+1:t+72]
PM10[t+1:t+72]
O3[t+1:t+72]
NO2[t+1:t+72]
```

Use direct multi-horizon prediction rather than recursively feeding model predictions back into itself whenever possible, because recursive error compounds over 72 hours.

# 13. Pollutant forecasts and AQI

## Predict pollutants first

Predict pollutant concentrations and derive AQI afterward.

Advantages:

- Preserves scientific meaning.
- Supports pollutant-specific alerts.
- Allows PM2.5, ozone and PM10 to be evaluated separately.
- Exposes model failure.
- Allows correct Indian AQI calculation.
- Supports health and policy interpretation.

Direct AQI prediction may be retained as a secondary auxiliary target, not the primary scientific output.



Indian AQI

CPCB’s method calculates pollutant sub-indices from concentration and breakpoint ranges using a segmented linear formula, then takes the maximum available pollutant sub-index. PM2.5 and PM10 use 24-hour averages, while O3 and CO use 8-hour averages in the standard method. [\[29\]](#) [\[30\]](#) [\[31\]](#) [\[32\]](#)

The general formula is:

$$I_p = \frac{I_{HI} - I_{LO}}{B_{HI} - B_{LO}}(C_p - B_{LO}) + I_{LO}$$

where  $C_p$  is the pollutant concentration and  $B$  and  $I$  are concentration and AQI breakpoints.

CPCB’s calculator lists, among others:

- PM2.5: 0–30 for AQI 0–50, 31–60 for 51–100, 61–90 for 101–200, 91–120 for 201–300, 121–250 for 301–400, 251–380 for 401–500.
- PM10: 0–50, 51–100, 101–250, 251–350, 351–430 and 431–510 across the six categories.
- O3: 0–50, 51–100, 101–168, 169–208, 209–748 and 749–1000. [\[33\]](#)

Use the current CPCB calculator/document as the implementation authority and version the breakpoint table.

14. Validation framework

Baselines

| Baseline               | Inputs                                                                | Purpose                                |
|------------------------|-----------------------------------------------------------------------|----------------------------------------|
| Persistence            | Last observed concentration                                           | Short-horizon lower bound              |
| Historical average     | Hour/day/season climatology                                           | Seasonal reference                     |
| Weather-only ML        | Weather and calendar                                                  | Tests meteorological explanatory power |
| Pollution + weather ML | Historical pollutant and weather                                      | Strong tabular baseline                |
| AI-only                | Neural sequence model                                                 | Tests deep-learning value              |
| Proposed hybrid        | Physical background + weather + pollution + fires + plume + inversion | Final system                           |

Metrics

Use:

- MAE.
- RMSE.
- Bias.
- Correlation.
- $R^2$ .
- MAPE only where concentrations are not near zero.
- Forecast skill relative to persistence.
- Spatial MAE.
- Station-level bias.
- AQI-category accuracy.
- Severe-episode recall.
- False-alarm rate.

- Prediction interval coverage.
- CRPS if a probabilistic model is implemented.

## **Lead times**

Report separately at:

- 1 hour.
- 6 hours.
- 12 hours.
- 24 hours.
- 48 hours.
- 72 hours.

## **Regime-stratified evaluation**

Create event labels for:

- Normal days.
- Strong inversion.
- Low PBL.
- Severe smog.
- Stubble-burning influence.
- Rainfall/clearing.
- Dust episodes.
- Ozone episodes.

## **Spatial validation**

Use:

1. Temporal holdout.
2. Spatial station holdout.
3. Event holdout.
4. Season holdout.

Never report only a random train-test split.

## **Required charts**

- Lead-time error curves.
- Station-wise error map.
- Actual-versus-predicted PM2.5.
- AQI category confusion matrix.
- Prediction interval coverage.
- Inversion-event performance.
- Fire-event performance.
- Forecast skill relative to persistence.
- 72-hour animation.

- Calibration curve for alert probabilities.

## 15. Existing platforms and opportunity gap

| Platform/system                 | Forecast               | Weather                    | Chemistry                   | Fire/plume                      | Inversion/PBL                      | Uncertainty                | Hyperlocal          |
|---------------------------------|------------------------|----------------------------|-----------------------------|---------------------------------|------------------------------------|----------------------------|---------------------|
| CPCB AQI                        | Current/monitoring     | Limited public context     | AQI pollutants              | No integrated plume             | Not central                        | No                         | Station-based       |
| SAMEER                          | Hourly AQI updates     | Limited                    | AQI dissemination           | No                              | No                                 | No                         | Station/city        |
| SAFAR                           | 1–3-day city forecasts | Yes                        | Atmospheric model           | Research-linked fire capability | Likely internal, not fully exposed | Limited public display     | Location-specific   |
| Delhi Air Quality Early Warning | Up to 72 hours         | Yes                        | Research forecasting system | Fire and regional processes     | Not fully dashboard-exposed        | Limited public display     | Delhi-focused       |
| CAMS                            | Global forecast        | Coupled atmospheric system | Strong                      | Global fire/emission inputs     | Model fields                       | Ensemble/product dependent | Coarse for NCR      |
| HYSPLIT                         | Trajectory/dispersion  | Meteorology-driven         | Optional chemistry          | Source trajectories             | Uses meteorology                   | Ensemble possible          | User-configured     |
| FLEXPART                        | Lagrangian transport   | Meteorology-driven         | Transport/deposition        | Strong for source influence     | Meteorology-dependent              | Ensemble possible          | User-configured     |
| IQAir/WAQI                      | Current AQI            | Limited                    | Aggregation                 | No coupled plume                | No                                 | Limited                    | Point/city          |
| OpenAQ                          | Data platform          | No forecast                | Aggregated measurements     | No                              | No                                 | No                         | Depends on stations |

SAFAR has historically provided location-specific current information and 1–3-day forecasts, and the Delhi air-quality early-warning system reports forecasts up to 72 hours. [\[34\]](#) [\[35\]](#) [\[36\]](#) [\[37\]](#)

### Opportunity gap

The strongest defensible gap is not “nobody forecasts Delhi air quality.” Existing systems already do that.

Your gap should be:

An auditable Delhi-NCR decision-support layer that combines station observations, weather forecasts, inversion/PBL diagnostics, satellite fire activity, plume transport, pollutant-level 72-hour predictions, uncertainty and plain-language explanations in one operational GIS interface.

That is stronger than claiming to replace SAFAR or CPCB.

## 16. Ranked USP candidates

### Top 10

| Rank | USP                               | Why meaningful                                                     |
|------|-----------------------------------|--------------------------------------------------------------------|
| 1    | Coupling evidence panel           | Shows WRF-Chem feedback-on versus feedback-off effects             |
| 2    | Inversion-aware forecast          | Explains why pollution may remain trapped                          |
| 3    | Fire-to-NCR transport risk        | Converts fire observations into wind-aligned influence probability |
| 4    | Pollutant-level AQI decomposition | Shows whether PM2.5, PM10 or O3 drives AQI                         |

| Rank | USP                                 | Why meaningful                                                                                       |
|------|-------------------------------------|------------------------------------------------------------------------------------------------------|
| 5    | Forecast uncertainty band           | Makes 72-hour output operationally honest                                                            |
| 6    | Source-region influence map         | Displays Punjab/Haryana/UP/Rajasthan transport sectors                                               |
| 7    | Recovery-window prediction          | Forecasts when rainfall, wind or PBL improvement may clear pollution                                 |
| 8    | Station-to-grid residual correction | Combines coarse physical fields with local observations                                              |
| 9    | Event replay mode                   | Reconstructs a documented historical smog episode                                                    |
| 10   | Decision simulator                  | Tests hypothetical fire reduction or traffic-emission scenarios without claiming exact policy impact |

## Full 30 USP list

| #  | USP                               | Value                                      | Difficulty | Data dependency          | Competitor position                 |
|----|-----------------------------------|--------------------------------------------|------------|--------------------------|-------------------------------------|
| 1  | Feedback-on/off experiment        | Demonstrates real coupling                 | High       | WRF-Chem                 | Rarely exposed in public dashboards |
| 2  | Inversion strength index          | Converts vertical stability into risk      | Medium     | ERA5/WRF profiles        | Differentiator                      |
| 3  | PBL pollution trap alert          | Identifies accumulation windows            | Medium     | PBL + PM                 | Strong differentiator               |
| 4  | Fire transport probability        | Avoids binary fire causation               | Medium     | FIRMS + winds            | Strong                              |
| 5  | FRP-weighted influence            | Better than fire counts                    | Medium     | FIRMS FRP                | Differentiator                      |
| 6  | Plume arrival window              | Estimates when source influence may arrive | Medium     | Trajectories             | Strong                              |
| 7  | Multi-pollutant AQI driver        | Explains AQI                               | Low        | Pollutant forecasts      | Useful, not novel alone             |
| 8  | Forecast uncertainty band         | Communicates risk                          | Medium     | Quantile/ensemble model  | Differentiator                      |
| 9  | Station reliability layer         | Shows data confidence                      | Low        | QC metadata              | Useful                              |
| 10 | Spatial holdout score             | Demonstrates generalisation                | Medium     | Station network          | Judge credibility                   |
| 11 | Event replay mode                 | Reproducible validation                    | Low        | Historical data          | Strong demo feature                 |
| 12 | Forecast skill versus persistence | Honest performance                         | Low        | Validation outputs       | Technical credibility               |
| 13 | Regional inflow/outflow map       | Shows boundary exchange                    | Medium     | Wind/transport           | Strong GIS feature                  |
| 14 | Fire-free counterfactual          | Scenario comparison                        | High       | Tagged/perturbed model   | Advanced; label as scenario         |
| 15 | Rainfall recovery forecast        | Supports decision timing                   | Medium     | Rain + PM                | Meaningful                          |
| 16 | Wind-sector alert                 | Simple transport explanation               | Low        | Wind fields              | Useful                              |
| 17 | Ozone regime alert                | Separates PM and photochemical risk        | High       | O3/NOx/VOCs proxies      | Differentiator                      |
| 18 | CAMS/WRF background blending      | Reduces coarse-field bias                  | Medium     | CAMS + local data        | Strong engineering                  |
| 19 | Multi-model ensemble              | Robust forecast                            | Medium     | Multiple models          | Strong                              |
| 20 | Conformal prediction              | Distribution-free intervals                | Medium     | Validation residuals     | Advanced                            |
| 21 | Forecast provenance               | Every map has source/time                  | Low        | Metadata                 | Government-friendly                 |
| 22 | Data freshness monitor            | Detects stale feeds                        | Low        | Ingestion logs           | Operationally valuable              |
| 23 | Alert explainability              | "Why alert now?"                           | Low        | Feature attribution      | Useful                              |
| 24 | District action cards             | Converts science to action                 | Medium     | District layer           | Impact-oriented                     |
| 25 | School exposure window            | Location/time advisories                   | Medium     | Forecast + user location | Requires careful health wording     |

| #  | USP                                   | Value                     | Difficulty | Data dependency     | Competitor position        |
|----|---------------------------------------|---------------------------|------------|---------------------|----------------------------|
| 26 | Construction/dust risk layer          | PM10-specific alerts      | Medium     | Land use/weather    | Useful                     |
| 27 | Roadside versus background comparison | Shows urban heterogeneity | Medium     | Station metadata    | Strong visual              |
| 28 | Mobile low-bandwidth mode             | Wider access              | Low        | API/cache           | Product differentiator     |
| 29 | Reusable city adapter                 | Mumbai/Pune expansion     | Medium     | Configurable domain | Scalability                |
| 30 | Scientific claim guardrails           | Prevents overclaiming     | Low        | Metadata/UI text    | Impresses technical judges |

## 17. Decision-support design

### Scientifically defensible features

#### Inversion warning

“High trapping potential: shallow PBL, weak winds and positive lower-atmosphere temperature gradient. Pollution accumulation risk is elevated.”

#### Fire influence warning

“Upwind fire activity detected in Haryana/Punjab. Current winds and trajectory ensemble indicate a transport-favourable pathway toward NCR.”

#### Recovery window

“Improved ventilation is likely after 18:00 due to increasing wind and PBL height. Confidence: medium.”

#### Pollutant driver

“Forecast AQI is driven primarily by PM2.5. O3 remains below the dominant sub-index.”

### Action cards

For officials:

- Review GRAP-relevant conditions.
- Inspect fire-affected source regions.
- Consider traffic/construction advisories.
- Monitor predicted deterioration window.

For schools:

- Outdoor-activity risk message.
- Indoor-air precautions.
- Avoid medical diagnosis or personalised health claims.

### What-if scenarios

Implement only as scenario analysis:

- Reduce fire emissions by 25%, 50% or 75%.
- Reduce traffic emissions by a chosen percentage.
- Increase ventilation using forecast weather.

- Compare baseline and intervention runs.

Do not call the output a measured policy impact unless it comes from a validated emissions perturbation experiment.

## 18. Source attribution

### What can be claimed

You can estimate:

- Upwind transport likelihood.
- Fire-enhanced transport risk.
- Local-versus-background residual.
- Meteorological accumulation conditions.
- Fire-related scenario sensitivity.
- Modelled regional contribution under explicit assumptions.

### What cannot be claimed without additional modelling

Do not claim exact:

- Traffic percentage.
- Industry percentage.
- Dust percentage.
- Stubble-burning percentage.
- Secondary aerosol percentage.

For defensible source apportionment, use:

- Tagged tracers in WRF-Chem.
- Chemical mass-balance methods.
- Receptor modelling.
- Validated emission inventories.
- Sensitivity experiments with uncertainty.

Phrase results as:

“The scenario with fire emissions removed reduced modelled PM<sub>2.5</sub> by X under this configuration.”

Not:

“Stubble burning caused X% of Delhi pollution.”

A 2021 study using MODIS fire counts and tagged tracers estimated a maximum daily mean stubble-burning contribution of approximately 30–35% during October–November 2021, demonstrating both the value and the model-dependence of such estimates. <sup>[38]</sup>

## 19. Uncertainty

Recommended implementation

Use a two-level system:

- 1. Quantile LightGBM for P10, P50 and P90 predictions.
- 2. Conformal calibration on a temporally held-out validation period.

Display:

```
PM2.5 forecast: 285 µg/m³
Expected range: 250–335 µg/m³
Confidence: medium
Main uncertainty: wind direction and fire-emission intensity
```

For severe events, use probability:

```
Probability of AQI > 300 in next 24 hours: 78%
```

Uncertainty should widen with:

- Longer lead time.
- Missing stations.
- Weak fire-data confidence.
- Weather-model disagreement.
- Unusual pollutant regimes.
- Spatial distance from observations.

20. GIS and dashboard

Recommended GIS stack

| Layer                   | Recommendation                       | Reason                                                  |
|-------------------------|--------------------------------------|---------------------------------------------------------|
| Database                | PostgreSQL + PostGIS                 | Mature geospatial storage and spatial queries           |
| API                     | FastAPI                              | Python scientific ecosystem and fast development        |
| Background jobs         | Celery + Redis or Prefect            | Scheduled ingestion and forecast workflows              |
| Scientific data         | xarray, NetCDF, Zarr, Dask           | Native gridded-data handling                            |
| Geospatial processing   | GeoPandas, Rasterio, GDAL            | Standard Python geospatial tools                        |
| Maps                    | MapLibre GL JS                       | Open, flexible, no mandatory proprietary map dependency |
| Lightweight alternative | Leaflet                              | Fast hackathon implementation                           |
| High-volume layers      | <a href="#">deck.gl/vector</a> tiles | Efficient animated and particle layers                  |
| Frontend                | React/TypeScript                     | Componentised dashboard                                 |
| Mobile                  | Kotlin/Android only for alerts       | Avoid duplicating the entire web GIS                    |

Major screens

## Overview

- Current AQI.
- 72-hour AQI range.
- Dominant pollutant.
- Confidence.
- Alert status.
- Data freshness.

## Forecast

- Hourly timeline.
- PM2.5, PM10, O3 and NO2 tabs.
- P10/P50/P90.
- Station and grid modes.
- Actual versus observed values for historical replay.

## Map

- AQI heatmap.
- Pollutant layer.
- Monitoring stations.
- Station health.
- NCR and district boundaries.

## Meteorology

- Wind vectors.
- Temperature.
- Humidity.
- Rainfall.
- PBL height.
- Pressure and stability.

## Inversion

- Vertical temperature profile.
- Inversion score.
- PBL height.
- Duration.
- Trapping explanation.

## Fire and plume

- FIRMS hotspots.
- FRP.
- Source-region totals.
- Trajectory ensemble.



- Arrival window.
- Confidence.

## Explainability

- Feature contribution.
- “Why did the forecast rise?”
- Fire, wind, PBL, lagged PM and humidity contributions.

## Historical replay

- Select event.
- Step through 72 hours.
- Compare observation, baseline and proposed forecast.
- Show fire, weather and inversion synchronously.

## Station details

- Raw and cleaned observations.
- Data completeness.
- Forecast accuracy.
- Pollutant trends.
- Station classification.

## Mobile recommendation

Do not build a full Android clone initially.

Build Android only for:

- Location-based AQI alert.
- Push notification.
- Saved locations.
- Offline last-known forecast.
- Simple pollutant and uncertainty card.
- Deep link into web dashboard.

The web application should contain scientific maps and analysis.

# 21. Backend architecture

```

CPCB / GFS / CAMS / ERA5 / FIRMS
    ↓
Scheduled ingestion workers
    ↓
Raw object storage + provenance
    ↓
QC and harmonisation
    ↓
PostgreSQL/PostGIS + Zarr
    ↓
Feature generation
    ↓

```

```

Physics/plume services
  ↓
ML forecast service
  ↓
Uncertainty and AQI service
  ↓
FastAPI
  ↓
React/MapLibre dashboard
  ↓
Android alerts

```

## Tables

```

stations
observations_raw
observations_clean
weather_grids
chemistry_grids
fire_detections
trajectories
inversion_profiles
features_hourly
forecasts
forecast_intervals
aqi_outputs
model_runs
data_sources
quality_flags
alerts

```

## API endpoints

```

GET /stations
GET /observations?station_id=&from=&to=
GET /forecast?lat=&lon=&horizon=72
GET /forecast/grid?pollutant=pm25&lead_hour=
GET /aqi
GET /fires?region=&hours=
GET /plume?source_region=&valid_time=
GET /inversion?location=
GET /pbl?location=
GET /explanation?forecast_id=
GET /validation?event_id=
GET /health/data-freshness

```

# 22. Compute feasibility

## Student prototype

| Component          | Minimum practical requirement    |
|--------------------|----------------------------------|
| Data ingestion     | 4 CPU cores, 8–16 GB RAM         |
| LightGBM/XGBoost   | Laptop or Colab CPU              |
| Small neural model | 16 GB RAM, optional T4 GPU       |
| PostGIS            | 2–4 CPU cores, 8 GB RAM          |
| Dashboard          | Low-cost cloud or college server |

| Component            | Minimum practical requirement                              |
|----------------------|------------------------------------------------------------|
| HYSPLIT/FLEXPART     | Separate CPU job, modest domain                            |
| WRF-Chem case        | Linux server, 32+ cores preferred, 64+ GB RAM desirable    |
| Storage              | 100–300 GB for prototype; more for multi-year gridded data |
| Operational forecast | Precomputed/cache-first architecture                       |

## WRF-Chem warning

A complete multi-year, multi-kilometre WRF-Chem training archive is not a realistic student task unless compute is provided by a research institution.

Use:

- One or two event hindcasts.
- Coarser domain for debugging.
- NetCDF compression.
- Subset variables.
- Zarr or cloud-optimised storage.
- Precompute only the features used by ML.

## Cloud choices

- Colab/Kaggle: notebooks and model prototypes, not dependable operations.
- AWS/GCP/Azure: suitable for final deployment, but set budgets and shutdown rules.
- College Linux server: best for WRF-Chem if available.
- Object storage: store raw files and model artifacts.
- Free tiers: adequate for dashboard proof-of-concept, not sustained WRF-Chem operations.

# 23. SIH demo plan

## Recommended event

Use the 6–13 November 2017 crop-residue-burning episode because published work specifically tested 72-hour forecasts, MODIS AOD assimilation, fire emissions and aerosol-radiation feedback.<sup>[6]</sup>

## Precompute

- ERA5 weather.
- CPCB observations.
- FIRMS fires.
- MODIS AOD where available.
- CAMS/MERRA-2 background.
- WRF-Chem hindcast.
- Feedback-on/off WRF-Chem runs.
- HYSPLIT/FLEXPART trajectories.
- ML forecasts.
- Validation metrics.

- Dashboard tiles.

## Live demo

1. Select the historical event.
2. Show initial AQI.
3. Animate fire hotspots.
4. Display wind and trajectory ensemble.
5. Show inversion and PBL collapse.
6. Display PM2.5 forecast.
7. Compare against CPCB observations.
8. Toggle coupled feedback.
9. Show uncertainty.
10. Trigger an alert.
11. Run a labelled fire-reduction scenario.
12. Explain what the official user should do.

## What must be labelled

- Historical replay.
- Hindcast.
- Operational forecast.
- Simulated scenario.
- Observation.
- Model output.
- Derived influence score.

Do not make a precomputed result appear to be a live WRF-Chem execution.

## 24. Claims to avoid

| Risky claim                     | Defensible replacement                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------|
| “100% accurate”                 | “Validated against persistence and ML baselines across lead times and events”           |
| “Exact source attribution”      | “Modelled source-region influence under stated assumptions”                             |
| “Real-time WRF-Chem”            | “Near-real-time hybrid forecast with offline WRF-Chem coupling evidence”                |
| “Street-level forecast”         | “Approximately 5-km grid with station-level correction”                                 |
| “FIRMS fire equals emissions”   | “FIRMS fire activity and FRP used as emission proxies”                                  |
| “Stubble burning caused X%”     | “Fire-removal sensitivity changed modelled concentration by X under this configuration” |
| “PBL observed directly”         | “Model/reanalysis-derived PBL height”                                                   |
| “Inversion measured everywhere” | “ERA5/WRF-profile inversion proxy calibrated where radiosonde data exist”               |
| “Forecast replaces CPCB/SAFAR”  | “Complementary decision-support and explainability layer”                               |
| “Works for all Indian cities”   | “Architecture is transferable after city-specific calibration”                          |
| “Satellite sees all fires”      | “Satellite detections have cloud, overpass and confidence limitations”                  |
| “AI understands chemistry”      | “AI learns residual relationships using physical and chemical features”                 |

## 25. Feasibility score

| Dimension               | Score / 10 | Reason                                                                               |
|-------------------------|------------|--------------------------------------------------------------------------------------|
| Data availability       | 7          | Many strong sources, but CPCB history and institutional data access are inconsistent |
| Scientific feasibility  | 8          | Established methods exist                                                            |
| Engineering feasibility | 8          | Hybrid architecture is practical                                                     |
| ML feasibility          | 9          | Strong tabular and ensemble approaches are available                                 |
| WRF-Chem feasibility    | 5          | Possible, but difficult for students                                                 |
| Compute feasibility     | 6          | Prototype feasible; full operations difficult                                        |
| Real-time feasibility   | 7          | Hybrid mode feasible                                                                 |
| Demo feasibility        | 9          | Historical replay can be compelling                                                  |
| SIH compliance          | 8          | Strong if coupling evidence is shown honestly                                        |
| Innovation              | 8          | Integration and explainability are meaningful                                        |
| Impact                  | 9          | Strong decision-support relevance                                                    |
| Scalability             | 8          | City adapter architecture is reusable                                                |
| Deployment feasibility  | 7          | Depends on data access and compute                                                   |

### Overall

**7.5/10 for the recommended hybrid system.**

The biggest technical risk is WRF-Chem configuration. The biggest data risk is reproducible historical and real-time access to CPCB and institutional feeds. The biggest competitive advantage is an auditable system connecting **forecast → mechanism → source-region influence → uncertainty → decision**.

## 26. Risk register

| Risk                             | Probability | Impact | Detection                    | Mitigation                                        | Backup                                     |
|----------------------------------|-------------|--------|------------------------------|---------------------------------------------------|--------------------------------------------|
| CPCB historical data unavailable | High        | High   | Download test                | Use current data + OpenAQ/OGD + documented subset | ERA5/CAMS plus available stations          |
| Portal/API changes               | Medium      | High   | Health checks                | Adapter interfaces and caching                    | Stored snapshots                           |
| WRF-Chem compilation failure     | High        | High   | Early smoke test             | Start immediately on Linux                        | Use published/precomputed case or WRF-only |
| Insufficient compute             | High        | High   | Runtime benchmark            | Coarse domain and one event                       | Physics-guided ML                          |
| Fire under-detection             | Medium      | Medium | Compare sensors              | Fuse MODIS/VIIRS and confidence                   | Region-level fire index                    |
| Emission uncertainty             | High        | High   | Sensitivity runs             | Calibrate with observations                       | Use proxy language                         |
| Satellite cloud gaps             | Medium      | Medium | Missingness flags            | Use multi-day window and reanalysis               | CAMS/MERRA-2                               |
| Station failure                  | Medium      | High   | Flatline/completeness checks | Spatial residual correction                       | Held-out station ensemble                  |
| Poor 72-hour accuracy            | Medium      | High   | Lead-time metrics            | Quantile ensemble and persistence comparison      | Alert only high-confidence events          |
| Model drift                      | Medium      | Medium | Rolling error monitor        | Retraining schedule                               | Seasonal recalibration                     |

| Risk                       | Probability | Impact | Detection                      | Mitigation                         | Backup                               |
|----------------------------|-------------|--------|--------------------------------|------------------------------------|--------------------------------------|
| Licensing issue            | Medium      | High   | Legal review                   | Use official terms and attribution | Replace restricted component         |
| Overclaiming coupling      | High        | High   | Judge questions                | Feedback-on/off experiment         | Explicitly label hybrid architecture |
| Weather forecast bias      | Medium      | High   | Compare forecast vs reanalysis | Bias correction                    | Multi-model weather ensemble         |
| Uncertainty miscalibration | Medium      | Medium | Coverage tests                 | Conformal calibration              | Wider conservative intervals         |
| GIS performance            | Medium      | Medium | Browser profiling              | Vector tiles and caching           | Static raster layers                 |
| API outage during demo     | Medium      | High   | Pre-demo test                  | Cache all assets                   | Fully replayable local demo          |

## 27. Development roadmap

Assume approximately 8–12 weeks for a serious build; compress only the demo layer for SIH.

| Phase                      | Deliverables                                | Technology                          | Difficulty | Team             |
|----------------------------|---------------------------------------------|-------------------------------------|------------|------------------|
| 1. Data acquisition        | Source registry, sample downloads, licenses | Python, requests, CDS/FIRMS clients | Medium     | Data + science   |
| 2. Data engineering        | Canonical hourly tables, QC                 | pandas, xarray, PostGIS             | High       | Data             |
| 3. Baselines               | Persistence, climatology, LightGBM          | LightGBM, sklearn                   | Medium     | ML               |
| 4. Weather/pollution model | Multi-pollutant forecast                    | LightGBM/XGBoost                    | High       | ML + science     |
| 5. Fire/plume              | FIRMS features and trajectories             | FIRMS, HYSPLIT/FLEXPART             | High       | Science + GIS    |
| 6. WRF-Chem                | One historical coupled case                 | Linux, WRF-Chem                     | Very high  | Atmosphere       |
| 7. Hybrid AI               | Residual correction and quantiles           | LightGBM/PyTorch                    | High       | ML               |
| 8. Validation              | Event and lead-time metrics                 | sklearn, matplotlib                 | Medium     | ML + science     |
| 9. GIS                     | Maps and time animation                     | PostGIS, MapLibre                   | High       | GIS/frontend     |
| 10. Dashboard              | Screens and API                             | FastAPI, React                      | High       | Backend/frontend |
| 11. Deployment             | Docker, monitoring, caching                 | Docker, Redis                       | Medium     | Backend          |
| 12. PPT/demo               | Evidence, claims, Q&A                       | Slides, recorded replay             | Medium     | Product/PPT      |

### Dependencies

- Data ingestion before modelling.
- Station registry before validation.
- WRF-Chem smoke test before promising it.
- Forecast model before dashboard integration.
- Validation before USP claims.
- Backend API before Android.

## 28. Six-person team

| Member                   | Responsibility                                                   |
|--------------------------|------------------------------------------------------------------|
| 1. ML/AI                 | Baselines, multi-horizon model, quantiles, explainability        |
| 2. Atmospheric modelling | WRF-Chem, inversion, PBL, chemistry interpretation               |
| 3. Data engineering      | CPCB, ERA5, CAMS, FIRMS ingestion and QC                         |
| 4. Backend/MLOps         | FastAPI, database, scheduler, Docker, monitoring                 |
| 5. GIS/frontend          | PostGIS, maps, plume animation, dashboard UX                     |
| 6. Android/product/PPT   | Mobile alerts, user personas, demo flow, claims and presentation |

Member 2 should define scientific requirements before Member 1 finalises features. Member 3 owns provenance and leakage prevention. Members 4 and 5 should agree on API schemas early.

## 29. Final technology stack

### Data

- Python.
- pandas and Polars for tabular processing.
- xarray and Zarr for multidimensional data.
- NetCDF4.
- Dask only when data volume requires it.

### Scientific computing

- WRF-Chem on Linux.
- WRF/WPS.
- HYSPLIT or FLEXPART.
- MetPy.
- GeoPandas.
- Rasterio.
- GDAL.

### ML

- LightGBM as the primary operational model.
- XGBoost as an independent baseline.
- PyTorch for TCN/ST-GNN experimentation.
- SHAP or native feature contributions.
- Conformal prediction library or custom calibration.

### Backend

- FastAPI.
- PostgreSQL/PostGIS.
- Redis.
- Celery or Prefect.

- Docker.
- Nginx or a managed reverse proxy.

## Frontend/GIS

- React + TypeScript.
- MapLibre GL JS.
- [deck.gl](https://deck.gl/) for plume and vector animation.
- Leaflet as a fallback.

## Mobile

- Kotlin.
- Android notifications.
- REST API.
- WorkManager for cached refreshes.

## Infrastructure

- Object storage for raw and model files.
- PostgreSQL backups.
- Prometheus/Grafana or a lightweight health dashboard.
- GitHub Actions for tests and Docker builds.

# 30. Cost analysis

## Prototype

| Item                   | Likely cost                                                       |
|------------------------|-------------------------------------------------------------------|
| Open-source software   | ₹0                                                                |
| ERA5/CAMS/FIRMS access | Usually no direct data charge, subject to registration/terms      |
| Local development      | ₹0 if existing laptops                                            |
| Cloud dashboard        | ₹0–₹2,000/month for small deployment                              |
| Database/storage       | ₹0–₹3,000/month depending on volume                               |
| GPU experiments        | ₹0–₹10,000 depending on provider and duration                     |
| WRF-Chem compute       | ₹0 if college server; potentially high on cloud                   |
| Map tiles              | ₹0 with self-hosted/open tile strategy; otherwise usage-dependent |
| Domain                 | Approximately ₹800–₹1,500/year                                    |
| Android distribution   | Optional; avoid unnecessary launch costs                          |

The major cost is not the web application. It is sustained WRF-Chem computation and storage.



# 31. Scalability

Create a city configuration file:

```
city: delhi_ncr
bbox: ...
grid_resolution: 0.05
stations: ...
source_regions: ...
weather_source: ...
pollutant_breakpoints: cpcb
seasonal_regimes: ...
```

Reusable components:

- Ingestion framework.
- Station schema.
- Forecast API.
- AQI service.
- Uncertainty service.
- GIS layers.
- Validation framework.
- Alert engine.
- Model registry.

City-specific components:

- Emission inventory.
- Station density.
- Meteorological regimes.
- Pollutant chemistry.
- Source regions.
- Calibration data.
- Fire and dust relevance.

Do not copy Delhi’s model weights directly to Mumbai or Bengaluru. Transfer learning and local recalibration are required.

# 32. Research literature to prioritise

Core papers and sources

| Work                                                                        | Relevance                                               |
|-----------------------------------------------------------------------------|---------------------------------------------------------|
| Grell et al., WRF-Chem foundational work                                    | Coupled meteorology–chemistry design                    |
| Emmons et al., MOZART                                                       | Gas-phase chemistry                                     |
| Fast et al., WRF-Chem aerosol/radiation applications                        | Feedback and aerosol modelling                          |
| 2024 ACP northern-India WRF-Chem evaluation                                 | Delhi/IGP PM2.5, meteorology, chemistry and validation  |
| Kumar et al. on chemical data assimilation during Delhi crop-burning season | 72-hour WRF-Chem forecast, MODIS AOD and fire emissions |
| 2021 stubble-burning contribution study                                     | Fire counts, tagged tracers and Delhi contribution      |

| Work                       | Relevance                                                  |
|----------------------------|------------------------------------------------------------|
| Chutia et al. 2024         | Aerosol–photolysis/radiative effects on Delhi PM2.5 and O3 |
| 2016 Delhi episode study   | Severe event and PM2.5 observation                         |
| CPCB National AQI report   | AQI formula and breakpoints                                |
| ERA5 documentation         | Reanalysis and vertical meteorology                        |
| CAMS documentation         | Atmospheric-composition background and forecast            |
| NASA FIRMS documentation   | Active-fire and FRP interpretation                         |
| NOAA HYSPLIT documentation | Trajectory and dispersion                                  |
| FLEXPART documentation     | Lagrangian transport and GPL software                      |

The 2024 WRF-Chem evaluation is particularly valuable because it reports both strengths and failures: good vertical temperature-gradient agreement, strong spatial aerosol correlations, but underestimation of peak PM2.5 and sensitivity to dust, night-time wind and missing chemistry.<sup>[4]</sup>

The crop-burning forecast study reported large improvements from MODIS AOD assimilation, aerosol-radiation feedback and fire emissions, with feedback contributing materially to bias reduction.<sup>[6]</sup>

### 33. Repositories and software

| Repository/system        | Purpose                              | License/access                            | Use                          |
|--------------------------|--------------------------------------|-------------------------------------------|------------------------------|
| WRF model GitHub/NCAR    | Numerical weather model and WRF-Chem | Public research software                  | Coupled hindcast             |
| WRF-Chem tools community | Emissions and preprocessing          | Official research tools                   | Required preprocessing       |
| NOAA HYSPLIT             | Trajectories and dispersion          | Use agreement; redistribution restriction | Plume layer                  |
| FLEXPART                 | Lagrangian transport                 | GPLv3                                     | Alternative plume engine     |
| xarray                   | Gridded scientific data              | Open source                               | ERA5/CAMS/WRF                |
| GeoPandas                | Vector geospatial processing         | Open source                               | Boundaries and stations      |
| Rasterio/GDAL            | Raster processing                    | Open source                               | Satellite and forecast tiles |
| PostGIS                  | Spatial database                     | Open source                               | GIS backend                  |
| MapLibre                 | Web maps                             | Open source                               | Frontend                     |
| <a href="#">deck.gl</a>  | Large-scale visualisation            | Open source                               | Plume animation              |

WRF downloads and current release information are maintained through the NCAR/MMM ecosystem and the WRF GitHub release workflow. FLEXPART’s GPLv3 status is documented in its scientific publication and community repositories.<sup>[26] [39] [40] [5]</sup>

### 34. Bookmark list

#### Must use

1. [CPCB real-time air-quality data](#) — live PM and station monitoring.<sup>[10]</sup>
2. [CPCB NCRAAQI portal](#) — NCR station and AQI access.<sup>[11]</sup>
3. [CPCB CAAQMS station list](#) — station metadata and coordinates.<sup>[9]</sup>
4. [CPCB AQI calculation document](#) — official calculation method.<sup>[30]</sup>
5. [ERA5 CDS](#) — historical weather and PBL.<sup>[14]</sup>
6. [NASA FIRMS](#) — fire and FRP.<sup>[18] [17]</sup>

7. [CAMS composition forecasts](#) — chemical background. <sup>[16]</sup>
8. [WRF-Chem official page](#) — coupled model. <sup>[5]</sup>
9. [WRF source/download](#) — model source. <sup>[39]</sup>
10. [NOAA HYSPLIT](#) — trajectory and dispersion. <sup>[23]</sup>

### Strongly recommended

- MERRA-2 aerosol reanalysis.
- MODIS AOD.
- TROPOMI/Sentinel-5P.
- GFS forecast data.
- NCR Planning Board boundaries.
- OpenStreetMap.
- FLEXPART.
- CPCB/DPCC station metadata.
- Radiosonde data where accessible.

### Optional

- EDGAR/HTAP/CEDS emissions.
- GFED/FINN/GFAS fire inventories.
- Land-use and population rasters.
- AERONET AOD.
- Dust emission products.
- Satellite cloud products.

### Research only

- Full multi-year WRF-Chem.
- Detailed VOC source inventories.
- Full chemical data assimilation.
- High-resolution source-apportionment chemistry.
- Neural ODE/PINN systems.

### Not worth the effort initially

- Full Android dashboard clone.
- Custom satellite fire detector.
- Training a Transformer before strong baselines.
- Street-level forecasts without dense observations.
- Exact intervention optimisation.
- Building a new AQI standard.
- Reproducing every pollutant if data completeness is poor.

## 35. Final recommended architecture

```
CPCB/DPCC/IMD observations
ERA5 historical weather
GFS forecast weather
CAMS/MERRA-2 composition
NASA FIRMS MODIS/VIIRS fires
MODIS AOD / TROPOMI
OSM and NCR boundaries
↓
Scheduled ingestion + provenance
↓
Immutable raw storage
↓
Quality control and station-health checks
↓
IST/UTC alignment and unit normalisation
↓
Station-to-grid background/residual fusion
↓
ERA5/WRF inversion and PBL diagnostics
↓
FIRMS FRP and source-region fire features
↓
HYSPLIT/FLEXPART transport trajectories
↓
Offline WRF-Chem case with aerosol-radiation feedback
↓
LightGBM/XGBoost multi-pollutant residual correction
↓
Quantile and conformal uncertainty
↓
PM2.5, PM10, O3, NO2 forecasts
↓
CPCB AQI calculation
↓
PostGIS and Zarr forecast storage
↓
MapLibre/deck.gl GIS engine
↓
React dashboard
↓
FastAPI alerts and Kotlin Android notifications
```

## 36. Final product priorities

### P0 — required

- Current CPCB station observations.
- 72-hour PM2.5 forecast.
- PM10 and O3 forecasts where data support them.
- CPCB AQI derivation.
- Delhi-NCR map.
- Weather forecast layer.
- PBL and inversion indicator.
- FIRMS fire map.
- Plume trajectory.

- Actual-versus-predicted validation.
- Uncertainty range.
- Data freshness and source labels.
- Historical replay.

#### **P1 — strong differentiators**

- WRF-Chem feedback-on/off comparison.
- Fire-to-NCR transport influence.
- Pollutant driver decomposition.
- Recovery-window forecast.
- Station reliability.
- Event-specific metrics.
- Decision cards.
- Quantile/confidence alerts.

#### **P2 — advanced**

- Tagged-tracer source attribution.
- Fire-reduction sensitivity.
- Traffic/construction scenarios.
- ST-GNN.
- Multi-model weather ensemble.
- TROPOMI trace-gas fusion.
- Automated anomaly detection.

#### **P3 — future**

- Full operational WRF-Chem.
- Multi-city transfer.
- Chemical data assimilation.
- Mobile personal exposure route planning.
- Policy optimisation.
- Public API.

## **37. PPT structure**

### **Slide 1 — Problem**

Message: Delhi pollution is a weather–chemistry feedback problem, not only an AQI problem.

Show:

- Delhi-NCR map.
- Severe-smog image or self-created AQI visual.
- 72-hour requirement.

Remember: “The challenge is coupled forecasting.”

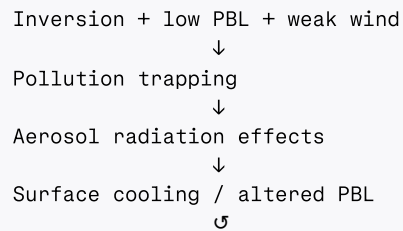
## Slide 2 — Existing gap

Show a comparison:

- CPCB: observations.
- Existing forecasts: limited public explanation.
- Generic ML: weak physics.
- Proposed: coupling, inversion, fire transport and uncertainty.

## Slide 3 — Scientific mechanism

Diagram:



## Slide 4 — Delhi-NCR influence geography

Show:

- Delhi-NCR boundary.
- Punjab/Haryana/UP/Rajasthan.
- Source-region arrows.
- Monitoring stations.

## Slide 5 — Data architecture

Show:

- CPCB.
- ERA5/GFS.
- CAMS.
- FIRMS.
- Satellite AOD.
- OSM/NCR boundaries.

## Slide 6 — Coupled modelling

Show:

- WRF-Chem hindcast.
- Feedback-on/off experiment.
- Operational hybrid path.

## Slide 7 — Fire and plume

Show:

- FIRMS hotspots.
- FRP.
- Wind-aligned trajectory ensemble.
- Arrival probability.

## Slide 8 — Inversion and PBL

Show:

- Vertical temperature profile.
- Inversion layer.
- PBL height.
- PM2.5 accumulation chart.

## Slide 9 — AI forecasting

Show:

```
Weather + pollution history + CAMS + fires + plume + inversion
      ↓
Multi-pollutant LightGBM
      ↓
Quantile uncertainty
```

## Slide 10 — AQI and explainability

Show:

- PM2.5/PM10/O3 forecasts.
- CPCB sub-index bars.
- Dominant pollutant.
- “Why forecast rose?”

## Slide 11 — Dashboard

Show your own screenshot:

- Map.
- Timeline.
- Fire layer.
- PBL/inversion card.
- Confidence band.

## Slide 12 — Validation

Show:

- Persistence versus proposed.
- 1-72 hour error curves.

- Historical event actual-versus-predicted chart.
- AQI category accuracy.

### Slide 13 — USPs

Show only the top five:

1. Feedback evidence.
2. Inversion-aware forecast.
3. Fire-to-NCR transport.
4. Uncertainty.
5. Decision-support timeline.

### Slide 14 — Impact

Show user personas:

- CPCB/official.
- Emergency manager.
- School.
- Transport authority.
- Public.

### Slide 15 — Feasibility and roadmap

Show:

- P0 implemented.
- WRF-Chem case.
- Hybrid operational system.
- Future city expansion.
- Honest limitations.

## 38. Visual assets

Create your own visuals wherever possible.

| Visual                  | Recommendation                                |
|-------------------------|-----------------------------------------------|
| Delhi-NCR map           | Generate from official boundary and OSM data  |
| Station map             | Generate from current CPCB registry           |
| Fire map                | Generate from FIRMS data with attribution     |
| Plume map               | Generate from HYSPLIT/FLEXPART output         |
| Inversion diagram       | Create scientifically accurate vector diagram |
| PBL diagram             | Create using WRF/ERA5 profile output          |
| AQI heatmap             | Generate from your own forecast grid          |
| Actual-versus-predicted | Generate from CPCB validation data            |
| Feedback comparison     | Generate from WRF-Chem on/off runs            |
| Dashboard screenshot    | Build and capture your own interface          |



| Visual               | Recommendation                                                       |
|----------------------|----------------------------------------------------------------------|
| Satellite image      | Use NASA/ESA product with attribution and terms check                |
| Architecture diagram | Create in Figma, <a href="https://draw.io">draw.io</a> or PowerPoint |

Avoid downloading random Delhi-smog photographs. They add little technical value and create licensing risk.

## 39. Judge Q&A preparation

### Scientific

#### 1. What is two-way coupling?

**Answer:** Weather affects chemical transport through wind, PBL, stability and radiation, while aerosols and gases affect radiation, photolysis and surface energy balance. We demonstrate this using a WRF-Chem feedback-on/off experiment; our operational model uses the resulting physics-informed features and observation correction.

**Evidence:** WRF-Chem shared-coordinate and radiation/photolysis coupling description. <sup>[5]</sup> <sup>[4]</sup>

#### 2. Why is Delhi not a closed box?

**Answer:** Air parcels and fire plumes cross district and national boundaries. Source regions can influence Delhi depending on wind direction, PBL and residence time.

#### 3. Why is PBL important?

**Answer:** A lower PBL concentrates emissions in a shallower layer and typically increases surface concentration, although the exact relationship depends on emissions, chemistry and vertical mixing.

#### 4. Is an FIRMS fire a PM2.5 emission?

**Answer:** No. It is an active-fire detection and FRP observation. We use it as a confidence-weighted activity proxy and do not equate it directly to PM2.5 mass.

#### 5. Can you prove stubble burning caused the peak?

**Answer:** Only through a tagged-tracer or sensitivity experiment. Our operational layer reports transport-favourable influence, while source percentages are not claimed without chemical transport evidence.

#### 6. Why forecast pollutants instead of AQI?

**Answer:** AQI is derived from pollutant concentrations. Pollutant-level forecasts preserve scientific meaning and enable pollutant-specific validation and explanations.

#### 7. Why include O3?

**Answer:** Ozone is a major AQI pollutant and responds to sunlight, NOx, VOC chemistry and transport. It is not simply a particulate proxy.

#### 8. Is your inversion index a standard government index?

**Answer:** No. It is a documented operational trapping-risk index derived from vertical temperature gradients, PBL, wind and stability, calibrated against historical distributions and radiosonde cases.

## 9. Why can WRF-Chem still be wrong?

**Answer:** Emissions, plume rise, chemistry, land-surface processes, boundary conditions and meteorological biases all affect results. Published Delhi/IGP evaluations show realistic patterns alongside peak-concentration errors.<sup>[4]</sup>

## 10. What is the primary season?

**Answer:** October–November, because crop burning, regional transport, inversion and low ventilation commonly overlap.

## ML and accuracy

## 11. Why LightGBM?

**Answer:** The feature set is heterogeneous and mostly tabular, including station lags, weather forecasts, fire indicators, plume scores and static geography. LightGBM is fast, strong on tabular data and practical for multi-horizon deployment.

## 12. Why not a Transformer?

**Answer:** We will test advanced sequence models only after leakage-safe baselines. A Transformer that beats no physical baseline but is difficult to validate is not automatically better.

## 13. How do you avoid leakage?

**Answer:** Every feature has an availability time. We use only information available at the forecast issue time, block temporal splits, hold out stations and simulate operational data latency.

## 14. What is your baseline?

**Answer:** Persistence, historical average, weather-only ML and pollution-plus-weather ML. The proposed system must beat persistence and demonstrate skill at multiple lead times.

## 15. What if your model is worse at 72 hours?

**Answer:** We report lead-time-dependent skill and widen uncertainty. The system can still provide useful early-warning probabilities even if point accuracy declines.

## 16. How will you handle rare severe episodes?

**Answer:** Use event-stratified evaluation, weighted training, regime features and separate severe-event recall metrics. We will not hide poor normal-day averages behind event failures.

## 17. How do you forecast unmonitored locations?

**Answer:** Use a gridded background field plus station-residual correction and report spatial uncertainty. We do not imply every grid cell is directly observed.

## 18. How do you evaluate uncertainty?

**Answer:** Prediction interval coverage, interval width, calibration and CRPS where probabilistic forecasts are available.

## WRF-Chem and deployment

### 19. Are you running WRF-Chem live?

**Answer:** Not as the only operational path. We use WRF-Chem for validated historical coupled experiments and physics-derived calibration, while the operational 72-hour path is a faster hybrid model. This is deliberate engineering, not a hidden limitation.

### 20. Why not run full WRF-Chem every three hours?

**Answer:** It requires robust Linux compute, emissions preprocessing, boundary conditions and failure recovery. A student prototype should not claim operational reliability without runtime evidence.

### 21. What is your fallback if WRF-Chem fails?

**Answer:** GFS/CAMS plus CPCB, FIRMS, plume trajectories and LightGBM residual correction. The system remains useful but labels the mode correctly.

### 22. Which chemistry mechanism?

**Answer:** We will start with a simpler validated configuration for debugging, then use MOZART/MOSAIC or the selected WRF-Chem chemistry setup for the research event. The final mechanism and emissions configuration will be documented.

### 23. What is the WRF-Chem domain?

**Answer:** A regional South Asia/IGP outer domain and Delhi-NCR nested domain for the research experiment. Exact grid spacing will be reported with runtime and sensitivity results, not invented in advance.

### 24. How do you initialise chemistry?

**Answer:** Use chemical initial/boundary fields from CAMS or appropriate chemical datasets, then combine anthropogenic, biogenic, dust and fire emissions with documented preprocessing.

### 25. What does feedback-on/off show?

**Answer:** The difference in meteorology and pollutant forecast between otherwise matched simulations estimates the effect of aerosol feedback under that configuration.

## Data

### 26. Is CPCB data openly available?

**Answer:** Current station and AQI data are publicly exposed through CPCB systems, but bulk historical access and endpoint behaviour must be tested and versioned. We maintain cached snapshots and source provenance. <sup>[8]</sup> <sup>[21]</sup> <sup>[10]</sup>

### 27. Why use ERA5 if it is not real-time?

**Answer:** ERA5 is for historical training, inversion derivation and hindcast analysis. Operational forecasts use forecast products such as GFS, with ERA5 used for evaluation and calibration.

### 28. Why use CAMS?

**Answer:** It provides regional chemical background fields and multiple atmospheric-composition variables that help fill gaps between sparse ground stations. It is not treated as a street-scale truth. <sup>[15]</sup> <sup>[16]</sup>

### **29. What are satellite limitations?**

**Answer:** Clouds, overpass time, retrieval gaps, column-versus-surface differences and spatial averaging. We use satellites as complementary evidence, not ground-station replacements.

### **30. How do you classify stations?**

**Answer:** Combine official metadata, coordinates, road density, land use, industrial proximity and manual review. The classification is a modelling feature, not an official regulatory label.

## **Accuracy and claims**

### **31. What accuracy do you guarantee?**

**Answer:** None in advance. We will report MAE, RMSE, bias, correlation, AQI-category accuracy and interval coverage by lead time and pollution regime.

### **32. What if the model does not beat persistence?**

**Answer:** That is a failure condition. We would investigate data leakage, forecast-weather error, event imbalance and emissions features rather than presenting a weak model as successful.

### **33. Can you claim hyperlocal forecasting?**

**Answer:** Only if we define resolution honestly. Our initial target is approximately 5-km grid plus station-level correction, not street-level exposure.

### **34. How does uncertainty help officials?**

**Answer:** It distinguishes a confident deterioration from a borderline forecast and supports risk-based decisions instead of treating a single point estimate as fact.

### **35. Why is 72-hour uncertainty important?**

**Answer:** Weather, fire detection, wind direction and chemical regime uncertainty compound with lead time. A point forecast without a range can mislead decision-makers.

## **GIS and product**

### **36. Why PostGIS?**

**Answer:** It stores station points, administrative boundaries, forecast cells, trajectories and spatial queries in one mature database.

### **37. Why MapLibre?**

**Answer:** It supports interactive vector maps without forcing dependence on a proprietary rendering stack.

### **38. Why not make Android first?**

**Answer:** The scientific dashboard is the core user interface. Android is limited to alerts and saved locations to avoid duplicating complex GIS functionality.

### **39. Who is the primary user?**

**Answer:** Environmental and emergency-management officials who need deterioration timing, source-region influence, confidence and action context. Public-facing views are simplified.

### **40. What is the decision output?**

**Answer:** Not only AQI, but the expected deterioration/recovery window, dominant pollutant, trapping conditions, transport evidence and confidence.

## **Innovation and feasibility**

### **41. What is actually novel?**

**Answer:** The defensible novelty is the integrated, auditable workflow connecting pollutant forecasts with inversion/PBL state, fire transport, feedback evidence, uncertainty and decisions—not any individual algorithm.

### **42. Do existing systems forecast Delhi air quality?**

**Answer:** Yes. SAFAR and Delhi early-warning systems already provide forecasts. We position our platform as a complementary explainable GIS and research-to-operations layer, not a replacement. <sup>[36]</sup> <sup>[34]</sup>

### **43. Why should judges believe the WRF-Chem claim?**

**Answer:** We show configuration files, domain details, input provenance, feedback-on/off results, runtime and validation plots. We do not rely on a logo or screenshot.

### **44. What is the minimum viable product?**

**Answer:** CPCB observations, GFS/ERA5 weather, FIRMS fire features, inversion/PBL diagnostics, pollutant-level LightGBM forecast, CPCB AQI, uncertainty and GIS dashboard, plus one reproducible WRF-Chem coupled case.

### **45. What will you not build?**

**Answer:** Full nationwide operational chemistry, street-level exposure, exact source apportionment, a complete Android clone and unvalidated policy optimisation.

### **46. How will it scale?**

**Answer:** Replace city configuration, station registry, source regions, boundary layers and calibration data while reusing ingestion, database, API, AQI, uncertainty and dashboard services.

### **47. How do you handle licensing?**

**Answer:** Track source terms and attribution in the data registry. Use open products where possible; do not redistribute restricted software such as HYSPLIT without permission. <sup>[25]</sup>

### **48. What if a source feed goes down?**

**Answer:** Health checks trigger fallback to cached observations, GFS/CAMS background and the last valid model run. The UI displays data freshness.

### **49. How do you distinguish observation from forecast?**

**Answer:** Every map and chart includes source, issue time, valid time, lead hour and status: observed, analysed, hindcast, forecast or scenario.

## 50. What would make your coupling fake?

**Answer:** If we only feed weather into ML, show trajectories, or manually alter PBL after prediction, that is not two-way coupling. We avoid this by running matched WRF-Chem feedback-on/off experiments and clearly separating the operational hybrid model from the coupled research model.

# 40. Final recommendation

## If I were evaluating this SIH proposal

I would build it — but only after modifying the claim and scope.

## Strongest reason

The problem combines high public impact, rich scientific content and a compelling visual demonstration: weather, inversion, PBL, fires, plume transport, pollutant forecasts and uncertainty.

## Biggest technical risk

WRF-Chem configuration and compute. Start with the WRF-Chem smoke test immediately, but do not make the entire product depend on it.

## Biggest data risk

CPCB historical and real-time access consistency. Cache data, version station lists and create source adapters.

## Biggest competitive advantage

An explainable source-to-receptor forecast:

“AQI is likely to deteriorate because PM2.5 is rising, the PBL is shallow, an inversion is persisting and upwind fire activity is aligned with transport toward NCR.”

## Minimum viable scientifically defensible solution

- CPCB station data.
- ERA5 historical and GFS forecast weather.
- CAMS background composition.
- FIRMS fires and FRP.
- Inversion and PBL diagnostics.
- LightGBM multi-pollutant forecast.
- CPCB AQI calculation.
- Quantile uncertainty.
- HYSPLIT/FLEXPART trajectory.
- One WRF-Chem feedback-on/off historical experiment.
- Validation against persistence and strong baselines.
- GIS dashboard.

## **Ideal solution**

- Operational multi-model ensemble.
- WRF-Chem forecast cycle.
- Tagged fire tracers.
- Chemical data assimilation.
- High-resolution nested domain.
- Calibrated uncertainty.
- Government workflow integration.
- Transferable city adapter.

## **Demonstrate**

- One known severe episode.
- Actual versus predicted PM2.5.
- Feedback-on versus feedback-off.
- Fire activity and plume path.
- Inversion/PBL timeline.
- Forecast range.
- AQI driver.
- Decision alert.

## **Do not attempt**

- Full nationwide system.
- Exact sectoral source apportionment.
- Street-level claims.
- Fully live WRF-Chem without runtime evidence.
- Neural Transformer before baseline validation.
- Full-feature Android replacement.
- Unsupported health or lives-saved claims.

# **BUILD THIS checklist**

## **Datasets**

- [ ] CPCB CAAQMS station registry.
- [ ] CPCB hourly/current observations.
- [ ] CPCB AQI breakpoints and calculator.
- [ ] ERA5 single-level hourly data.
- [ ] ERA5 pressure-level temperature, humidity and wind.
- [ ] GFS operational weather forecasts.
- [ ] CAMS composition fields.
- [ ] NASA FIRMS VIIRS and MODIS fires.
- [ ] FIRMS FRP and confidence.

- ☐ MODIS AOD for selected events.
- ☐ TROPOMI NO<sub>2</sub>/CO/O<sub>3</sub> where practical.
- ☐ NCR and district boundaries.
- ☐ OSM roads, industries and land-use proxies.
- ☐ One emissions inventory.
- ☐ One historical severe pollution event.

## Software

- ☐ Python.
- ☐ pandas/Polars.
- ☐ xarray, NetCDF, Zarr.
- ☐ GeoPandas, Rasterio, GDAL.
- ☐ LightGBM and XGBoost.
- ☐ PyTorch only for advanced model testing.
- ☐ WRF/WPS/WRF-Chem on Linux.
- ☐ HYSPLIT or FLEXPART.
- ☐ PostgreSQL/PostGIS.
- ☐ FastAPI.
- ☐ Redis plus Celery/Prefect.
- ☐ Docker.
- ☐ React/TypeScript.
- ☐ MapLibre/deck.gl.
- ☐ Kotlin Android only for alerts.

## Models

- ☐ Persistence baseline.
- ☐ Historical-average baseline.
- ☐ Weather-only ML.
- ☐ Pollution-plus-weather ML.
- ☐ Fire/plume/inversion ML.
- ☐ Multi-pollutant LightGBM.
- ☐ Quantile forecasts.
- ☐ Conformal calibration.
- ☐ WRF-Chem historical feedback-on/off case.
- ☐ HYSPLIT/FLEXPART trajectory ensemble.

## Modules

- ☐ Data ingestion.
- ☐ Provenance registry.
- ☐ Station quality control.
- ☐ Temporal alignment.
- ☐ Spatial fusion.



- ☐ Inversion detection.
- ☐ PBL processing.
- ☐ Fire feature generation.
- ☐ Plume transport.
- ☐ Forecast inference.
- ☐ AQI conversion.
- ☐ Uncertainty.
- ☐ Validation.
- ☐ Alert logic.
- ☐ Data freshness monitoring.

## Screens

- ☐ Overview.
- ☐ Live AQI.
- ☐ 72-hour pollutant forecast.
- ☐ AQI heatmap.
- ☐ Station map.
- ☐ Weather layer.
- ☐ Inversion/PBL panel.
- ☐ Fire activity.
- ☐ Plume tracking.
- ☐ Explainability.
- ☐ Uncertainty.
- ☐ Historical replay.
- ☐ Station details.
- ☐ Alerts.
- ☐ Data provenance.

## Experiments

- ☐ Persistence comparison.
- ☐ Weather-only comparison.
- ☐ Pollution-weather comparison.
- ☐ Fire-feature ablation.
- ☐ Inversion-feature ablation.
- ☐ CAMS-background ablation.
- ☐ Feedback-on/off WRF-Chem.
- ☐ Fire-emission sensitivity.
- ☐ Spatial holdout.
- ☐ Temporal holdout.
- ☐ Severe-event evaluation.
- ☐ Rainfall-clearing evaluation.

- ☐ Prediction interval calibration.

## Graphs and maps

- ☐ Actual-versus-predicted PM2.5.
- ☐ Lead-time MAE.
- ☐ Forecast skill versus persistence.
- ☐ AQI category confusion matrix.
- ☐ Station error map.
- ☐ Inversion/PBL versus PM2.5 chart.
- ☐ Fire FRP and Delhi PM2.5 timeline.
- ☐ Feedback-on/off comparison.
- ☐ Forecast uncertainty band.
- ☐ Delhi-NCR station map.
- ☐ FIRMS fire map.
- ☐ Plume trajectory map.
- ☐ AQI forecast animation.
- ☐ Pollutant-driver chart.

## PPT assets

- ☐ Official/verified SIH26082 wording.
- ☐ Delhi-NCR influence map.
- ☐ Scientific feedback diagram.
- ☐ Architecture diagram.
- ☐ Data-source logos with attribution.
- ☐ WRF-Chem configuration evidence.
- ☐ Fire/plume demonstration.
- ☐ Inversion/PBL visualization.
- ☐ Actual-versus-predicted validation.
- ☐ Uncertainty screenshot.
- ☐ Dashboard screenshot.
- ☐ Top-five USP slide.
- ☐ Honest limitations slide.
- ☐ Roadmap and feasibility slide.
- ☐ Judge Q&A evidence sheet.

✱

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